



DEPARTMENT OF
COMPUTER SCIENCE

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BSc in Computer Science and Engineering

PLATFORM FOR DISSERTATION/PROJECT PROPOSAL AND MANAGEMENT

Dissertation Plan
MASTER IN COMPUTER SCIENCE AND ENGINEERING
SPECIALIZATION IN PROGRAMMING LANGUAGES AND SOFTWARE SYSTEMS

NOVA University Lisbon

Draft: January 10, 2026



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ABSTRACT

Regardless of the language in which the dissertation is written, usually there are at least two abstracts: one abstract in the same language as the main text, and another abstract in some other language.

The abstracts' order varies with the school. If your school has specific regulations concerning the abstracts' order, the NOVAthesis L^AT_EX (`novathesis`) (L^AT_EX) template will respect them. Otherwise, the default rule in the `novathesis` template is to have in first place the abstract in *the same language as main text*, and then the abstract in *the other language*. For example, if the dissertation is written in Portuguese, the abstracts' order will be first Portuguese and then English, followed by the main text in Portuguese. If the dissertation is written in English, the abstracts' order will be first English and then Portuguese, followed by the main text in English. However, this order can be customized by adding one of the following to the file `5_packages.tex`.

```
\ntsetup{abstractorder={<LANG_1>,...,<LANG_N>}}  
\ntsetup{abstractorder={<MAIN_LANG>={<LANG_1>,...,<LANG_N>}}}
```

For example, for a main document written in German with abstracts written in German, English and Italian (by this order) use:

```
\ntsetup{abstractorder={de={de,en,it}}}
```

Concerning its contents, the abstracts should not exceed one page and may answer the following questions (it is essential to adapt to the usual practices of your scientific area):

1. What is the problem?
2. Why is this problem interesting/challenging?
3. What is the proposed approach/solution/contribution?
4. What results (implications/consequences) from the solution?

Keywords: One keyword · Another keyword · Yet another keyword · One keyword more
· The last keyword

RESUMO

Independentemente da língua em que a dissertação está escrita, geralmente esta contém pelo menos dois resumos: um resumo na mesma língua do texto principal e outro resumo numa outra língua.

A ordem dos resumos varia de acordo com a escola. Se a sua escola tiver regulamentos específicos sobre a ordem dos resumos, o template (L^AT_EX) *novathesis* irá respeitá-los. Caso contrário, a regra padrão no template *novathesis* é ter em primeiro lugar o resumo *no mesmo idioma do texto principal* e depois o resumo *no outro idioma*. Por exemplo, se a dissertação for escrita em português, a ordem dos resumos será primeiro o português e depois o inglês, seguido do texto principal em português. Se a dissertação for escrita em inglês, a ordem dos resumos será primeiro em inglês e depois em português, seguida do texto principal em inglês. No entanto, esse pedido pode ser personalizado adicionando um dos seguintes ao arquivo `5_packages.tex`.

```
\abstractorder(<MAIN_LANG>):={<LANG_1>,...,<LANG_N>}
```

Por exemplo, para um documento escrito em Alemão com resumos em Alemão, Inglês e Italiano (por esta ordem), pode usar-se:

```
\ntsetup{abstractorder={de={de,en,it}}}
```

Relativamente ao seu conteúdo, os resumos não devem ultrapassar uma página e frequentemente tentam responder às seguintes questões (é imprescindível a adaptação às práticas habituais da sua área científica):

1. Qual é o problema?
2. Porque é que é um problema interessante/desafiante?
3. Qual é a proposta de abordagem/solução?
4. Quais são as consequências/resultados da solução proposta?

Palavras-chave: Primeira palavra-chave · Outra palavra-chave · Mais uma palavra-chave · A última palavra-chave

CONTENTS

Glossary	xiii
Acronyms	xv
Symbols	xvii
1 Introduction	1
1.1 Context and Background	1
1.2 Problem Statement	1
1.3 Objectives	3
1.4 Methodology	3
1.4.1 Your Time is Precious	4
1.4.2 Recognition	5
1.5 Getting Started	5
1.5.1 Using Overleaf	5
1.5.2 Using a Local L ^A T _E X Installation Local	7
1.6 Getting Help	7
1.6.1 Suggestions, Bugs and Feature Requests	8
1.7 Donors	8
1.8 Disclaimer	8
2 State of the art and Technologies	9
2.1 Backend Architectures and BaaS	9
2.2 Data Persitence: SQL vs. NoSQL	9
2.3 Frontend Frameworks and User Experience	9
2.4 AI-Assisted Development (Vibe Coding)	9
3 Requirements and Current System Analysis	11
3.1 Analysis of the Legacy System	11
3.2 Requirementes Gathering	11

3.3	Use Cases and Stakeholders	11
4	Solution Proposal	13
4.1	Proposed System Architecture	13
4.2	Process Modelling (BPMN)	13
4.3	Data Model (ERD)	13
4.4	API and Component Design	13
4.5	Work Plan	13
5	NOVAtesis Template <i>User's Manual</i>	15
5.1	Introduction	15
5.2	Quick Start	15
5.2.1	With a Local L ^A T _E X Installation	15
5.2.2	With a Remote Cloud-based Service	17
5.3	Folder and Files	17
5.4	Customizing the <i>novathesis</i> template	19
5.4.1	Options in <code>1_novathesis.tex</code>	19
5.5	How to Write Using L ^A T _E X	24
5.6	Example glossary, acronyms, and symbols	24
6	A Short L^AT_EX Tutorial with Examples	25
6.1	Document Structure	25
6.2	Dealing with Bibliography	25
6.3	Inserting Tables	25
6.4	Importing Images	25
6.5	Floats, Figures and Captions	25
6.6	Text Formatting	27
6.7	Generating PDFs from L ^A T _E X	27
6.7.1	Generating PDFs with <code>pdflatex</code>	27
6.7.2	Dealing with Images	28
6.7.3	Dealing with Citations	28
6.7.4	Footnotes	28
6.7.5	Tables	28
6.7.6	Figures	29
6.8	Equations	29
6.9	Test for algorithms	29
7	Adding Support to a New School (work in progress)	33
	Bibliography	35
	Appendices	

A	<i>NOVAt</i>hesis covers showcase	37
A.1	A section here	41
B	Appendix 2 Lorem Ipsum	45
Annexes		
I	Annex 1 Lorem Ipsum	51

LIST OF FIGURES

1.1	The <code>novathesis</code> project web page in GitHub.	6
1.2	NOVAthesis template in Overleaf.	6
1.3	The NOVAthesis Project page in GitHub.	7
6.1	A figure with two sub-figures!	26
6.2	Bitmap image (JPG/PNG)	30
6.3	Vectorial image (PDF)	31
6.4	Exemplo de utilização de <i>subbottom</i>	32

LIST OF TABLES

5.1	The folders and files (top level).	17
5.2	The configuration files (Config folder).	18
6.1	Test results summary.	28

GLOSSARY

computer An electronic device which is capable of receiving information (data) in a particular form and of performing a sequence of operations in accordance with a predetermined but variable set of procedural instructions (program) to produce a result in the form of information or signals. This is a test that adds a citation [2] to the glossary! (*p. 24*)

ACRONYMS

aaa	acronym aaa (<i>p.</i> 24)
aab	acronym aab (<i>p.</i> 24)
aba	acronym aba (<i>p.</i> 24)
abbrev	abbreviation of a longer text (<i>p.</i> 24)
bbb	acronym bbb (<i>p.</i> 24)
FCT	NOVA School of Science and Technology (<i>p.</i> 15)
novathesis	NOVATHESIS L ^A T _E X (<i>pp.</i> i, iii, ix, 5–8, 15–18)
xpto	and extension of a xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto (<i>p.</i> 24)

SYMBOLS

μ Mu (*p. 24*)

π the numerical value of pi (*p. 24*)

r the radius of a circle (*p. 24*)

INTRODUCTION

1.1 Context and Background

The transition of students from academic life to the professional market or advanced research is a centerpiece of the university experience, essentially enabled through three different distinct paths at the Department of Informatics (DI) of NOVA FCT:

- Undergraduate Internships: Primarily offered by external companies to provide students with their first professional experience.
- Master's Dissertations: Scientific thesis typically conducted within the university, focusing on academic research under the supervision of professors.
- Engineering Projects: Practical projects proposed by external companies or the Department of Informatics itself, allowing Master's students to apply advanced engineering principles to real-world projects.

Managing those processes involves a complex coordination of multiple stakeholders: students seeking opportunities aligned with their skills and interests, professors overseeing academic quality, and external companies providing practical challenges. Within this ecosystem, a reliable digital platform is essential to act as a central hub for communication, proposal submission, and student placement.

1.2 Problem Statement

At DI@NOVA FCT, the management of internships, dissertations and engineering projects involves a multi-stage lifecycle that includes proposal submission, clarification and approval processes, making proposals available to students, report/dissertation submission, and providing documentation to juries. Currently the department relies on a legacy platform to support some of these activities; however this system presents several critical issues that hinder the efficiency of the department.

The most significant limitation is that the current platform is not being utilized for the Master's degree processes - including both scientific dissertations and engineering projects. At the moment it is exclusively used to manage undergraduate internships, meaning that the more complex workflows associated with Master's students are handled through manual, or non-standardized methods. This creates a lack of centralized data and increases the administrative burden on coordinators, professors and the secretariat.

Furthermore, even within its current scope, the legacy system suffers from several functional and technological gaps:

- **Communication Gaps and Reliability:** The current method for "Call for Proposals" is manual, requiring coordinators to send individual emails to companies, advertising the opportunity to send proposals, for each edition. There are no safeguards to ensure that these emails follow specific deliverability rules, often resulting in messages being flagged as spam. Additionally, the system lacks the flexibility to create and edit email templates, preventing coordinators from customizing communication for different editions or context.
- **Inefficient Feedback Loops:** Although companies can submit proposals, there is no integrated mechanism for coordinators to provide direct feedback on those proposals for clarification. This forces corrections to be handled outside the platform, whereas a centralized system would allow companies to correct and resubmit proposals based on coordinator comments.
- **Stale Company Data:** The existing repository contains a large amount of outdated information, including companies that no longer exist or have changed their contacts or other information. There is a lack of a "validation" status to ensure only active and verified companies can send proposals.
- **Manual Protocol Management:** For new companies that have never participated in previous editions, the process of signing the required protocols is still entirely manual. The platform should automate delivery and tracking of these legal documents as soon as a new company registers.
- **Outdated User Experience (UX):** The interface is not intuitive for external stakeholders, making it difficult for companies to register, manage their employees' accounts, or monitor the status of their proposals in real-time.

In conclusion, the necessity for a new platform arises from the need to unify all academic degrees under a single digital infrastructure, replacing manual tracking with a secure, automated system that ensures data integrity and seamless experience for all stakeholders.

1.3 Objectives

The primary objective of this dissertation is to design and develop a modernized web-based platform that unifies and automates the management and entire lifecycle of undergraduate internships, Master's dissertations, and engineering projects for DI @ NOVA FCT. The projects aims to transition from a fragmented, manual system to a centralized digital ecosystem.

To achieve this goal, the following specific objectives have been defined for the initial version:

- **Process Unification:** Integrate the lifecycles of undergraduate internships and Master's degree pathways (both scientific and engineering-focused) into a single platform, replacing the current manual methods.
- **Workflow Automation:** Implement automated procedures for the "Call for Proposals student seriation (ranking), and the validation of proposals.
- **Enhanced Communication and Feedback:** Establish direct feedback loops within the platform, allowing coordinators to request clarifications on proposals and ensuring companies can resubmit corrected versions.
- **Document and Protocol Management:** Automate the delivery, tracking, and storage of legal protocols, student CVs, and recommendation letters.
- **Administrative Oversight:** Develop advanced administrative tools, including "superuser" access for student support, detailed system logs for auditing, and validation workflow for new company registrations.

1.4 Methodology

This project will follow an Agile/Scrum methodology, characterized by iterative development, frequent feedback loops to ensure the final platform aligns with the university's complex requirements. The work plan is structured into several phases, prioritizing rapid prototyping followed by a transition to a high-performance custom architecture.

To achieve this goal, the following specific objectives have been defined for the initial version:

1. **Requirement Gathering and Analysis:** A thorough analysis of the legacy system and current manual processes was conducted to identify functional and non-functional requirements. This phase includes the definition of user roles (Admin, Coordinator, Student, Professor, Company, Secretariat) and their respective use cases.
2. **System Modeling and Design:** Utilizing BPMN (Business Process Model and Notation) to map complex workflows and ERD (Entity-Relationships Diagrams) to design the database schema. This phase also involves planning the system architecture.

3. **AI-Assisted Prototyping (Vibe Coding):** The initial implementation phase will leverage Vibe Coding tools, such as Lovable, Figma Make or Google AI Studio, to rapidly generate the frontend and core functional interfaces. These tools excel at creating React-based user interfaces through iterative natural language prompting. During this stage, Supabase will be utilized as Backend-as-a-Service (BaaS) to provide an immediate infrastructure for authentication, database management, and initial data structures via SQL. This approach allows for the rapid fulfillment of most functional requirements while maintaining a live, testable version of the platform.
4. **Architectural Transition and Manual Development:** Once the prototype reaches a stable state and meets the primary functional requirements, the project will transition from AI-generated code to manual development. This move is essential to refine the User Experience (UX), optimize performance, and implement custom features that go beyond the limitations of AI-driven boilerplate.
5. **Custom Backend Implementation:** To support the department's more complex business logic - such as the specialized student seriation (ranking) algorithms and robust administrative auditing - the system will transition to a dedicated custom backend, likely using Spring Boot. While Supabase is effective for rapid prototyping, a Spring Boot architecture offers superior domain modeling, type safety and maintainability through its layered structure (Controller, Service, Repository). This transition ensures that the platform is not only fast to develop initially but also scalable and reliable for long-term university use.
6. **Verification and Validation:** The platform will be tested against the initial requirements, ensuring that automated emails follow specific deliverability rules avoiding being flagged as spam. Additionally, the seriation logic will be rigorously validated to ensure it correctly handles student placements and matching according to pre-defined criteria.
7. **Documentation and Evaluation:** There will be a continuous documentation of the architecture decisions, providing detailed justifications for technical choices such as the preference for SQL over NoSQL for structured academic data, and other technical choices.

1.4.1 Your Time is Precious

Did you learn how to drive by sitting by the wheel and throwing your car into the road? Most probably you did take your time *learning the rules* and *practicing* first! Likewise, it is not wise to throw yourself at the task of writing a thesis/dissertation in L^AT_EX without seriously considering the following recommendation!

If you are going to spend zillions of hours writing your thesis/dissertation using the **novathesis** $\text{\LaTeX}$ template (or some other $\text{\LaTeX}$ template), be wise and spend a couple of hours learning how to use it properly by reading its manual. And then, be even wiser, and spend a few more hours learning some $\text{\LaTeX}$. I am sure that the time you are investing now will pay itself countless times before you submit your thesis/dissertation.

— João Lourenço

1.4.2 Recognition

The **novathesis** template was born in 1996, and what you see now accumulates to many many hundreds (thousands?!) of working hours, unpaid and stolen from family and friends. This work is available to the community under the $\text{\LaTeX}$ Project Public License v1.3c, which means you are entitled to use it for free and change it at your will. However, if you decide to use this template to write your thesis/dissertation, **be fair to the developers** and:

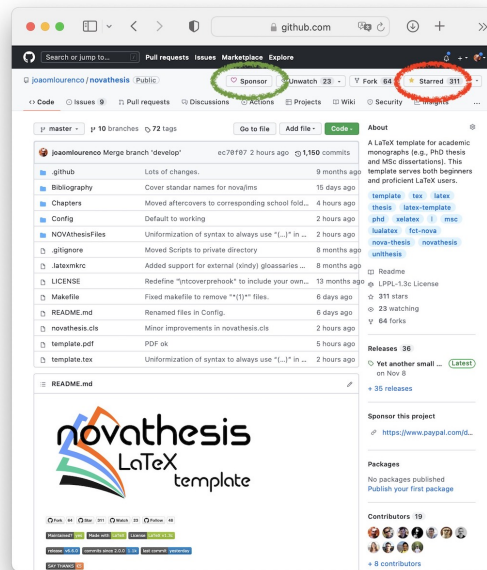
1. Cite the **novathesis** manual [11] in a place of your choice (e.g., in the *Acknowledgments*) of your thesis/dissertation with “`\cite{novathesis-manual}`”. If you cite it this way, the correct entry will be added automatically to your bibliography (no need to worry with the necessary BibTeX entry, as it will be added automatically);
2. Go to the project web page in GitHub and give the project a star (marked with a red ellipse at the top-right in Figure 1.1); and
3. Make a donation by visiting the **novathesis** project page and clicking in the button marked with a green ellipse at the top-center in Figure 1.1). Alternatively, just click [HERE](#) and your browser will be directed to the right page.

1.5 Getting Started

The template provides an *easy to use* setting for you to write your thesis/dissertation in $\text{\LaTeX}$:

- Select your school;
- Fill your thesis metadata (title, research field, etc) in the file “`template.tex`”;
- Create your thesis/dissertation contents using the files in folder “`Chapters`”; and
- Process using you favorite $\text{\LaTeX}$ processor (pdf $\text{\LaTeX}$, Xe $\text{\LaTeX}$ or Lua $\text{\LaTeX}$).

1.5.1 Using Overleaf

Figure 1.1: The **novathesis** project web page in GitHub.

Overleaf is a collaborative cloud-based LaTeX editor used for writing, editing and publishing scientific documents. Like “Google Docs”, for $\text{\LaTeX}$ users. You can edit and compile your $\text{\LaTeX}$ source on the cloud, without installing software in your own computer, and, much like *Google Docs*, you can share your document with others users and everybody can edit the same file at the same time (this may be dangerous).

If you do not have an account in Overleaf, you must create one first.

Once you have an account, please access the **novathesis** template in Overleaf and select the green button *Open as Template* (see Figure 1.2).

Please notice that the version currently available in Overleaf (v7.1.18) is slightly outdated (current version is v7.5.5). A new version (v7.1.29) will be submitted to Overleaf soon. Until then, please:

1. Download the latest version from the GitHub repository as a Zip file.
2. Login to your favorite LaTeX cloud service. I recommend Overleaf but there are alternatives (these instructions apply to Overleaf and you'll have to adapt for other providers).
3. In the menu select: **New project** → **Upload project**.
4. Upload the zip file.
5. Select “template.tex” as the main file.

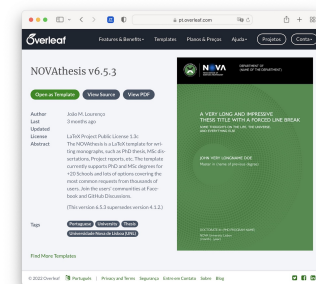


Figure 1.2: NOVathesis template in Overleaf.

6. Let Overleaf compile the document.

Notice that you need a (student) subscription to compile the **novathesis** template in Overleaf, otherwise your compilation will always time out.

1.5.2 Using a Local L^AT_EX Installation Local

First of all, start by installing L^AT_EX in your computer. There are two main distributions, Mik_T_EX and T_EXLive, and both of them are available for the 3 most popular Operating Systems: Linux, macOS and Windows.

Be aware that a full installation of Mik_T_EX or T_EXLive will take near 5 GB of hard disk space. So, think twice before installing the full distribution. See the **novathesis** Wiki for the list of packages required to compile the template.

Once you have L^AT_EX up and running, remember to install a good L^AT_EX text editor. I recommend you to take a look at this post in the `tex.stackexchange.com` site. If you want a quick and dirty recommendation, try TeXStudio.

Now, you must access the **novathesis** repository in GitHub, select the green button *Code* and then *download* (or *clone*) the template. You will always get the latest version of the template (currently v7.5.5 from 2025-11-14).

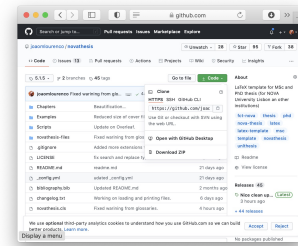


Figure 1.3: The NOVAt_hesis Project page in GitHub.

1.6 Getting Help

No! You don't have to use this template to write your thesis. You don't even have to use L^AT_EX. However, writing a thesis is serious stuff, and which tool you shall use to write it is not a decision to make lighthearted.

L^AT_EX is hard enough by itself. This template aims at making your life easier, but not easy. If you choose to use this template to write your thesis, you are very welcome. However, don't expect me to provide you help with L^AT_EX. Look for help with your friends (you have some friends, don't you?), or search the web, or try even to read some book(s) on L^AT_EX. In the end you will certainly find the experience rewarding.

When you come to the point of *"How do I do this with the novathesis template?"*, remember. . .

1. To check the **novathesis** wiki and have some hope! :D
2. Google is your best friend.
3. Search the GitHub Discussions page for a question related to yours. *If and only if* you don't find one, then post your own question in English please!

4. Search the NOVAtheis Facebook group for a question related to yours. *If and only if* you don't find one, then post your own question in either Portuguese or English, at your preference.

When you post your own question, remember to **always** state the **novathesis** version number you are using and referring to.

Please do not attempt to contact me directly (email, Messenger, etc). . .
I WILL NOT REPLY!

1.6.1 Suggestions, Bugs and Feature Requests

Help: If you just need some help, see above Section 1.6.

Suggestion: Do you have a suggestion/recommendation? Please add it to the wiki and help other users!

Bug: Did you find a bug? Please open an issue. Thanks!

New Feature: Would you like to request a new feature (or support of a new School)? Please open an issue. Thanks!

1.7 Donors

The list of *Donnors* is available in the **novathesis** Project page.

1.8 Disclaimer

Although the **novathesis** template is endorsed by some Schools (e.g., linked from NOVA FCT web site), the **novathesis** template **this not an official template** for any School.

The **novathesis** template exists to make your life easier and we do our best to make it compliant to the supported (+25) Schools' regulations but, in the end of the line, you and only you are accountable for both the look and the contents of the document you submit as your thesis/dissertation.

STATE OF THE ART AND TECHNOLOGIES

- 2.1 Backend Architectures and BaaS**
- 2.2 Data Persistence: SQL vs. NoSQL**
- 2.3 Frontend Frameworks and User Experience**
- 2.4 AI-Assisted Development (Vibe Coding)**

REQUIREMENTS AND CURRENT SYSTEM ANALYSIS

- 3.1 Analysis of the Legacy System
- 3.2 Requirements Gathering
- 3.3 Use Cases and Stakeholders

SOLUTION PROPOSAL

- 4.1 Proposed System Architecture**
- 4.2 Process Modelling (BPMN)**
- 4.3 Data Model (ERD)**
- 4.4 API and Component Design**
- 4.5 Work Plan**

NOVATHESIS TEMPLATE *USER'S MANUAL*

This manual is outdated and must be revised!

5.1 Introduction

This Chapter describes how to use the NOVAtHesis L^AT_EX (`novathesis`) template. It is assumed that you have a working of L^AT_EX, either local (in your own computer) or remote (in Overleaf), and that you were able to generate a PDF for the default configuration of the template: a PhD thesis for NOVA School of Science and Technology (FCT).

5.2 Quick Start

5.2.1 With a Local L^AT_EX Installation

Follow these steps to get started with a local L^AT_EX installation:

1. Download L^AT_EX. There are two major L^AT_EX distributions — MikT_EX and T_EXLive — that share lots of similarity, and L^AT_EX documents are portable between them. This means that, for most users, both systems are equally usable.

T_EX-Live is maintained by (La)T_EX developers and is certainly the best distribution you may install in your computer: However, the default distribution will take more than 5 GB on your hard disk. . . so, if you are not short on disk space, install T_EXLive!

MikT_EX will, by default, install only a minimal set of packages. The extra/additional packages will be installed on the fly. Installing packages on the fly is useful if disk space is limited, but has its own caveats in the longer term. Definitely choose MikT_EX if you're short on disk space.

Which one to download? There are pros and cons for both distributions so it is essentially a question of where does your heart fall first! Mine falls to T_EXLive, but yours can fall elsewhere! :D

2. Install $\text{\LaTeX}$. Installation of $\text{\LaTeX}$ is as hard as installing any other software. Just do your best and you will certainly succeed.
3. Update your $\text{\LaTeX}$ installation using the *TeXLiveUtility* program of the MikTeX console.
4. Download the **novathesis** template by either:

- Cloning the GitHub repository with

```
git clone --depth=1 https://github.com/joaomlourengo/novathesis.git
```

or

- Downloading the latest version from the GitHub repository as a Zip file.

5. Download additional School specific files if applicable:

Universidade do Minho (UMINHO) download the required *NewsGotT* font files

from <https://github.com/joaomlourengo/novathesis-extras/raw/main/Fonts/NewsGotT.zip>

then unzip the file and copy the 3 font files

`"n015002t.ttf", "n015003t.ttf", and "n015006t.ttf"`

to the folder

`"NOVAthesisFiles/FontStyles/Fonts"`.

Escola Superior de Enfermagem do Porto (ESEP) download the required *Calibri*

font files from <https://github.com/joaomlourengo/novathesis-extras/raw/main/Fonts/Calibri.zip>

then unzip the file and copy the 4 font files

`"Calibri.ttf", "Calibrib.ttf", "Calibrii.ttf", and "Calibriz.ttf"`

to the folder

`NOVAthesisFiles/FontStyles/Fonts`.

6. Compile the document with you favorite LaTeX processor (pdfLaTeX, XeLaTeX or LuaLaTeX).

- The main file is named `"template.tex"`, but you are free to rename it as you please.
- Either load the main file in your favorite LaTeX text editor and press the appropriate (*magic*) button to generate a PDF file, or open a terminal and compile it with `"latexmk -pdf template"`. If you use a $\text{\LaTeX}$ text editor, please notice that the **novathesis** template uses `"biber"` and not `"bibtex"` to process the bibliography, which means that most probably you have to open

the *Editor Preferences* and somewhere (depending on the Editor you are using) change “bibtex” to “biber”.

- Notice that, due to the external font sets used, pdfL^AT_EX will not work for both UMINHO and ESEP, and you have to use either X_YL^AT_EX (“latexmk -pdfxe template”) or LuaL^AT_EX (“latexmk -pdflua template”).

7. Edit the files in the “Config” folder. See Table 5.2.

8. Recompile de document.

9. And you’re done with a beautifully formatted thesis/dissertation! :D

5.2.2 With a Remote Cloud-based Service

Follow these steps to get started with a remote L^AT_EX installation:

- Download the latest version from the GitHub repository as a Zip file.
- Login to your favorite LaTeX cloud service. I recommend Overleaf but there are alternatives. These instructions apply to Overleaf and you’ll have to adapt for other providers.
- In the menu select New project → Upload project.
- Select “template.tex” as the main file.
- Follow from Step 5 above in Section 5.2.1 (With a Local L^AT_EX Installation).

5.3 Folder and Files

The novathesis template is organized into many files and folders. At the main level it includes the following files and folders listed in Table 5.1.

Table 5.1: The folders and files (top level).

Name	Type	Access	Contents
template.tex			The main template file. You need to <i>compile</i> this file with one of pdfL ^A T _E X, X _Y L ^A T _E X, or LuaL ^A T _E X to obtain the PDF file (“template.pdf”). I recommend the usage of the “latexmk” command or, if you use a UN*X-like OS, you may use “make” (and the ggiven “Makefile”).
Config			Configuration files. Please customize your template by changing the files in this folder!

Chapters			Examples of document contents, including Chapters, Appendices, Annexes, Abstracts, Glossaries, Lists of Symbols, etc. Replace them with your own.
Bibliography			Where all your bibliography files should be located. You may have has many bibliography files as you want.
template.pdf			A possible result of applying pdf $\LaTeX$ to the “template.tex” file. The look and feel of the document will depend on the parametrization/configuration (e.g., School) of this template.
novathesis.cls			The main class file.
NOVAthesisFiles			Additional files for the novathesis template. This is where all the juice is so, unless you are a TeXmagician, don't mess up with the files and folders inside this folder.

Table 5.2: The configuration files (Config folder).

Name	Type	Access	Contents
0_memoir.tex			Options specific for the memoir class. <i>Don't touch this file unless you know what you are doing!</i>
1_novathesis.tex			The main configuration file for the template, e.g., select the document type, the school, the used languages, etc.
2_biblatex.tex			Select how your citations and bibliographic references will be printed. The default is numbers inside square brackets, e.g. [11], but you can change it to other formats, such as author-year, e.g., Lourenço (2021).
3_cover.tex			Configure cover contents (e.g., author's name, thesis/dissertation title, author, advisers, committee, etc)
4_files.tex			Select which files shall be included in the document as chapters, appendices, annexes, etc. . .
5_packages.tex			User's customization, such as loading additional packages and declare user defined commands.

6_list_of.tex			Configure the lists to be printed (table of contents, list of figures, list of tables, list of listings, etc). <i>Don't touch this file unless you know what you are doing!</i>
9_nova_fct.tex			Configurations specific to NOVA FCT. <i>Otherwise ignored.</i>
9_ulisboa_fmv.tex			Configurations specific to ULISBOA FMV. <i>Otherwise ignored.</i>
9_ulisboa_ist.tex			Configurations specific to ULISBOA IST. <i>Otherwise ignored.</i>
9_uminho.tex			Configurations specific to UMINHO (all schools). <i>Otherwise ignored.</i>

5.4 Customizing the *novathesis* template

The *novathesis* template can be customized by editing the files in the Config folder.

5.4.1 Options in 1_novathesis.tex

5.4.1.1 Most Relevant Options (1_novathesis.tex)

doctype=OPT <i>phd, phdprop, phdplan, msc, mscplan, bsc, plain</i>	The type of the document. phd → PhD thesis (← <i>default</i>). phdprop → PhD thesis proposal (for NOVA FCT). phdplan → PhD thesis plan. msc → MSc thesis. mscplan → MSc thesis plan. bsc → BSc report. plain → Other report.
--	---

school=OPT

*nova/fct, nova/fcsh, nova/ims,
nova/ims/mcsig, nova/ims/mgt,
nova/ensp, nova/itqb/green,
nova/itqb/gray, ulisboa/ist, ulisboa/fcul,
ulisboa/fmv, uminho/eaad, uminho/ec,
uminho/ed, uminho/eeg, uminho/eeng,
uminho/elach, uminho/emed,
uminho/epsi, uminho/ese, uminho/i3bs,
uminho/ics, uminho/ie, iscteiu/eta,
ips/ests, ipl/isel, ipl/isel/meb, ulht/deisi,
ulht/mge, other/esep*

Selection of the university and of the school (and degree variant).

⇒ *Default: school=nova/fct*

This option changes the typesetting of the document to some specific School formatting and layout, like covers, margins, fonts, paragraph spacing and indentation, etc.

docstatus=OPT

draft, provisional, final

The current status of the document.

working → Working version (⇐ *default*).
provisional → Version for submission.
final → Final version.

lang=OPT

en, pt, de, es, fr, gr, it

The main language for the document.

en → English (⇐ *default*).
pt → Portuguese.
de → German.
es → Spanish.
fr → French.
gr → Greek.
it → Italian.

media=OPT

screen, paper

The target media for the PDF.

screen → No empty/white pages (⇐ *default*).
paper → Empty/white pages are added when necessary.

print/webography=OPT

User defined title

Generate a separate bibliography for @online references.

⇒ *Default: print/webography=undefined*

If undefined, the @online references are listed in the main bibliography. If defined, the @online references will be printed in a separate bibliography titled as given in the option.

color/links=OPT

Color name

The color for the hyperlinks (URLs, cross references, citations).

⇒ *Default: color/links=DarkBlue*

The valid color names as listed in “xcolor” manual, the “svgname” color set.

color/gls=OPT <i>Color name</i>	The color for the glossary managed hyperlinks (glossary, symbols, etc). $\Rightarrow$ <i>Default: color/gls=Black</i> The valid color names as listed in “xcolor” manual, the “svgname” color set.
print/index=OPT <i>true,</i> <i>false</i> ($\Leftarrow$ default)	Print the (words) index at the end of the document. Print the index (in Portuguese <i>Índice Remissivo</i>).

5.4.1.2 Less Relevant Options (1_novathesis.tex)

abstractorder=OPT $L_0 = \{L_1, L_2, \dots, L_n\}$	Forces the abstracts languages and order for documents in language L_0. $\Rightarrow$ <i>Default: abstractorder={en={en,pt}} for english</i> $\Rightarrow$ <i>Default: abstractorder={L={L,en}} for lang L</i> L_i is a two-letters language code from the set of valid language codes, following ISO 3166-1 (alfa-2).
lang/extra=OPT $\{L_1, L_2, \dots, L_n\}$	List of additional languages are used in the document besides the main language and those used in the abstracts (above). $\Rightarrow$ <i>Default: lang/extra={}</i> L_i is a two-letters language code from the set of valid language codes, following ISO 3166-1 (alfa-2).
gnumberlist=OPT <i>true,</i> ($\Leftarrow$ default) <i>false</i>	Shall the glossary entries list the page numbers where those entries are used? (Like a reverse index!)
numberallpages=OPT <i>true,</i> <i>false</i> ($\Leftarrow$ default)	Shall all the pages (except cover) be numbered?
tocintoc=OPT <i>true,</i> <i>false</i> ($\Leftarrow$ default)	Shall table of contents be listed in the table of contents?
tocintoc=OPT <i>true,</i> <i>false</i> ($\Leftarrow$ default)	Shall a second cover page be forced? If the contents for the second page are not defined, the second cover will be a replica of the first cover.

print/committee=OPT

true,
false

Shall the evaluation committee be printed?

⇒ *Default: print/committee=true if docstatus=final*
⇒ *Default: print/committee=false otherwise*

print/frontmatter=OPT

true, (⇐ default)
false

Shall the front matter be printed?

Set to false to generate a PDF with only the cover and the book chapters.

print/statement=OPT

true,
false (⇐ default)

Shall the honor/originality statement be printed?

print/copyright=OPT

true, (⇐ default)
false

Shall the copyright message be printed?

print/timestamp=OPT

true,
false

Shall a timestamp (with the PDF generation date/time) be printed in the cover?

⇒ *Default: print/timestamp=true if docstatus=working*
⇒ *Default: print/timestamp=false otherwise*

style/url=OPT

default, (⇐ default)
same

Use the same (main) font in URLs?

style/font=OPT

arial, bookman, calibri, erewhon,
kieranhealy, kpfonts, libertine, newpx,
newsgott, scholax

Which font set to use in the document?

arial → Use 'arial' font. Requires Xe^LA_TE_X or Lua^LA_TE_X.
bookman → Use 'bookman' font.
calibri → Use 'calibri' font. Requires Xe^LA_TE_X or Lua^LA_TE_X.
erewhon → Use 'erewhon' font.
kieranhealy → Use 'kieranhealy' font.
kpfonts → Use 'palatino' font.
libertine → Use 'libertine' font.
newpx → Use 'palatino' font. (⇐ *default*)
newsgott → Use 'newsgott' font. Requires Xe^LA_TE_X or Lua^LA_TE_X.
scholax → Use 'scholax' font.

style/chapter=OPT

See list on the side!

Which chapter style to use in the document?

Besides the standard memoir chapter styles (default, section, article, reparticle, hangnum, companion, demo, bianchi, bringhurst, brotherton, chappell, culver, dash, demodemoell, ger, lyhne, madsen, pedersen, southall, thatcher, veelo, verville, crosshead, dowding, komalike, ntglie, tandh, wilsondob), the customized list of chapter styles below is also available.

bar → Use ‘bar’ chapter style. ($\Leftarrow$ *default*)

bar-compact → Use ‘bar-compact’ chapter style.

bluebox → Use ‘bluebox’ chapter style.

compact → Use ‘compact’ chapter style.

elegant → Use ‘elegant’ chapter style.

fmv → Use ‘fmv’ chapter style.

hansen → Use ‘hansen’ chapter style.

ist → Use ‘ist’ chapter style.

ist2 → Use ‘ist2’ chapter style.

pedersen → Use ‘pedersen’ chapter style.

lang/cover=OPT*en, pt, de, es, fr, gr, it***The main language for the cover.** $\Rightarrow$ *Default: The same as the main language.***lang/copyright=OPT***en, pt, de, es, fr, gr, it***The main language for the copyright message.** $\Rightarrow$ *Default: The same as the main language.***spine/layout=OPT***no, full, trim***Print the “book spine” at the end of the document?** $\Rightarrow$ *Default: ‘trim’ if docstatus=final* $\Rightarrow$ *Default: ‘no’ otherwise*

no → do not print the book spine.

full → print the book spine in a full page.

trim → print and trim the page to the width of the book spine.

spine/width=OPT $\text{\LaTeX}$ dimension**Force the width of the “book spine”.** $\Rightarrow$ *Default: The “natural width”.*

The default width for the book spine will be the width of the number of pages of the document if printed in standard paper ($80\text{g}/\text{m}^2$).

debug=OPT*cover, spine***Activate debug mode for cover and/or book spine.** $\Rightarrow$ *Default: debug={}*

5.5 How to Write Using $\LaTeX$

Please have a look at Chapter 6, where you may find many examples of $\LaTeX$ constructs, such as Sectioning, inserting Figures and Tables, writing Equations, Theorems and algorithms, exhibit code listings, etc.

5.6 Example glossary, acronyms, and symbols

This is the first occurrence of an abbreviation: abbreviation of a longer text (abbrev). And now the second occurrence of the same abbreviation: abbrev. And a new acronym with capital letter: And extension of a xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto xpto (xpto) and reused xpto. Let's also use a few other acronyms such as acronym aaa (aaa), acronym aab (aab), acronym aba (aba), acronym bbb (bbb) and xpto. In geometry, the area enclosed by a circle of radius r is πr^2 . Here the Greek letter π is equal to the ratio of the circumference of any circle to its diameter. Lets add "computer" to the glossary! Be carefull with mathematical symbols in acronyms, please see the definition of μ .

A SHORT L^AT_EX TUTORIAL WITH EXAMPLES

This Chapter aims at exemplifying how to do common stuff with L^AT_EX. We also show some stuff which is not that common! ;)

Please, use these examples as a starting point, but you should always consider using the *Big Oracle* (aka, Google, your best friend) to search for additional information or alternative ways for achieving similar results.

6.1 Document Structure

6.2 Dealing with Bibliography

Citing something online [4, 7, 9].

6.3 Inserting Tables

6.4 Importing Images

6.5 Floats, Figures and Captions

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

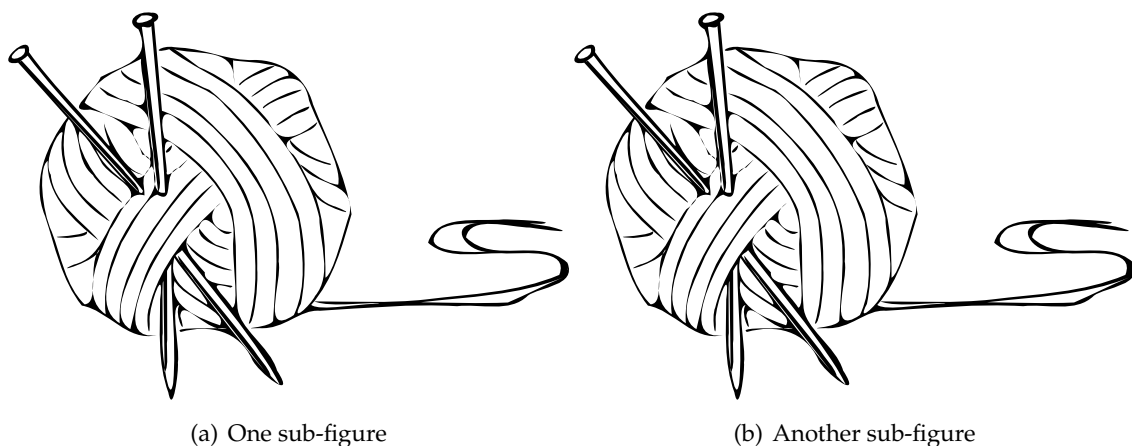


Figure 6.1: A figure with two sub-figures!

And this is a small text that references the Figure 6.1 and its Subfigures 6.1(a) and 6.1(b).

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan

eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

6.6 Text Formatting

6.7 Generating PDFs from $\text{\LaTeX}$

6.7.1 Generating PDFs with `pdflatex`

You may create PDF files either by using `latex` to generate a DVI file, and then use one of the many DVI-2-PDF converters, such as `dvipdfm`.

Alternatively, you may use `pdflatex`, which will immediately generate a PDF with no intermediate DVI or PS files. In some systems, such as Apple, PDF is already the default format for $\text{\LaTeX}$. I strongly recommend you to use this approach, unless you have a very good argument to go for `latex + dvipdfm`.

A typical pass for a document with figures, cross-references and a bibliography would be:

```
$ pdflatex template
$ bibtex template
$ pdflatex template
$ pdflatex template
```

You will notice that there is a new PDF file in the working directory called `template.pdf`. Simple :)

Please note that, to be sure all table of contents, cross-references and bibliographic citations are up-to-date, you must run `latex` once, then `bibtex`, and then `latex` twice.

6.7.2 Dealing with Images

You may process the same source files with both `latex` or `pdflatex`. But, if your text include images, you must be careful. `latex` and `pdflatex` accept images in different (exclusive) formats. For `latex` you may use EPS ou PS figures. For `pdflatex` you may use JPG, PNG or PDF figures. I strongly recommend you to use PDF figures in vectorial format (do not use bitmap images unless you have no other choice).

6.7.3 Dealing with Citations

Para fazer citações, deverá usar-se a chave da referência no ficheiro BibTeX. Se for uma única referência [2], usar um “~” para ligar o `\cite{...}` à palavra que o precede (...referência~\cite{Artho04}). Caso queira fazer múltiplas citações [12, 13, 6], deverá agrupá-las dentro de um único `\cite{...}`.

Note que o ficheiro de bibliografia pode ter tantas entradas quantas quiser. Apenas aquelas cuja chave seja referenciada no texto é que serão incluídas na listagem de bibliografia.

6.7.4 Footnotes

Footnotes¹ will be numbered and shown in the bottom of the page.

6.7.5 Tables

The Table 6.1 illustrates some important concepts associated with table construction:

- i) Do not use vertical lines;
- ii) The caption should be above the table;
- iii) Use the macros `\toprule`, `\midrule` and `\bottomrule` to make the top, inner and bottom horizontal lines, respectively.

Table 6.1: Test results summary.

Test	Anomalies	Warnings	Correct	Categories	Missed
Connection [3]	2	2	1	C	1
Coordinates’03 [1]	1	4	1	2B, 1C	0
Local Variable [1]	1	2	1	A	0
NASA [1]	1	1	1	—	0
Coordinates’04 [2]	1	4	1	3C	0
Buffer [2]	0	7	0	2A, 1B, 2C, 2D	0
Double-Check [2]	0	2	0	1A, 1B	0
StringBuffer [8]	1	0	0	—	1
Account [14]	1	1	1	—	0

¹This is a simple footnote.

Jigsaw [14]	1	2	1	C	0
Over-reporting [14]	0	2	0	1A, 1C	0
Under-reporting [14]	1	1	1	—	0
Allocate Vector [10]	1	2	1	C	0
Knight Moves [3]	1	3	1	2B	0
Total	12	33	10	5A, 6B, 10C, 2D	2

6.7.6 Figures

The images inserted in the document must be of good quality, preferably in vector format (vector PDF) and not in *bitmap* (PNG, JPG, etc.). *bitmap* images (Figure 6.2) do not scale well and have negative effects on the quality of your document. On the other hand, *vector* images Figure 6.3 scale as much as necessary without degrading the quality of the image.

You should only use *screenshots* for your plots, charts, etc, if you absolutely have no other alternative. Instead of generating a *screenshot*, try using a virtual PDF printer and printing to a PDF file. As a general rule, you will get a vector PDF. Even if your PDF contains images, they will always be of higher or equal quality than what you would get with a *screenshot*.

To combine several figures into a single one... You can then reference the set as Figure 6.4 or the sub-figures separately as 6.4(a) and 6.4(b).

6.8 Equations

LaTeX is a powerful tool for writing in a mathematical style. It allows you to insert formulas into the text, such as this: $ax^2 + bx + c = 0$. It also allows formulas to be highlighted on a separate line and centered on the page.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

or numbererd

$$e = mc^2 \tag{6.1}$$

which can latter be referenced as equation 6.1

6.9 Test for algorithms

Uncomment the algorithms source below and add the following to file “5_packages.tex”

```
\usepackage{algorithm2e}
\RestyleAlgo{ruled}
```

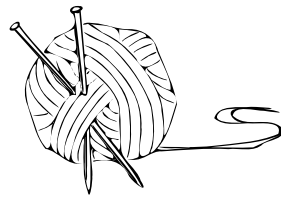
and uncomment



Figure 6.2: Bitmap image (JPG/PNG)



Figure 6.3: Vectorial image (PDF)



(a) Novelo de lã



(b) Tempestade com neve

Figure 6.4: Exemplo de utilização de *subbottom*

```
\ntaddlistof{listofalgorithms}
```

in file “8_list_og.tex”.

ADDING SUPPORT TO A NEW SCHOOL (WORK IN PROGRESS)

My advice to customize the **novathesis** template to another School/University/Department/Degree is to browse the existing supported degrees to find one that is *close enough*, and depart from there!

The multitude of layouts supported by the **novathesis** template is based in a three-tier naming scheme, separated by slashes: University / School / Department-or-Degree. This three-tier naming scheme is also reflected in a three-tier directory (folder) structure in: `<project_root>a/NOVAthesiFiles/Schools/....` For example:

```
...
|
+-- nova
|   +-- Images
|   +-- fct
|   |   \-- Images
|   +-- ims
|   |   \-- Images
|   ...
|
\-- uminho
    +-- Images
    +-- ea
    |   \-- Images
    +-- ec
    |   \-- Images
    ...
```

The directory `uminho` contains the customization for all Schools of Universidade do Minho. This university is an example of the case where the regulations are defined at

University level and all the schools apply the same thesis layout and organization. So, the all the customization is done in the file `uminho/uminho-defaults.ldf`, except the definition of the name and logo of each individual school.

As another example, the directory `nova` contains the customization for all Schools from NOVA University Lisbon. This university grants a lot of freedom in the definition of the thesis layouts. In some cases, they are defined at the School level (e.g., NOVA FCT), while in some other cases they are defined separately for each degree (e.g., NOVA IMS).

1. Try all the already supported schools and check which one is closer to your needs;
 - a) Edit `Config/1_novathesis.tex` and near line 28 uncomment the line with key `\ntsetup{school=<SOMETHING>};`
 - b) For each school supported (see the comment), replace `<SOMETHING>` with the school name, e.g., make it `\ntsetup{school=ulisboa/fmv}`
 - c) Recompile and check the document. Particularly, check the cover layout, the front-page (second cover) layout, the front-matter contents, the bibliography style;
 - d) Repeat for the next school, until you find one close enough.
- 2.

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NOVATHESIS COVERS SHOWCASE

This Appendix shows examples of covers for some of the supported Schools. When the Schools have very similar covers (e.g., all the schools from Universidade do Minho), just one cover is shown. If the covers for MSc dissertations and PhD thesis are considerable different (e.g., for NOVA FCT and UMinho), then both are shown.

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APPENDIX 2 LOREM IPSUM

This is a test with citing something [5] in the appendix.

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ANNEX 1 LOREM IPSUM

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